

Feature-based Learning of Tonal Processes at a Sub-symbolic Neural Level

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Abstract

This study shows that tonal processes are learnable from tonal features, which in turn can be learned from the raw continuous input values of a sub-symbolic network in the framework of Gradient Symbolic Computation (Smolensky and Goldrick, 2016b) (GSC). As a case study, we examine an intriguing H-raising process in Baatonum, where a H tone raises to Top before a morpheme boundary followed by a Low tone. Our results show the advantages of modeling feature-based learning of tonal phenomena in GSC, where features are represented by neurons at the sub-symbolic level and constraints on their interaction by directly interpretable neural-level connections. We show that this kind of modeling can capture tonal phenomena like these in a way that would be difficult for purely symbolic models.

1 Introduction

We focus on a puzzling tonal interaction in Baatonum (Gur, Benin), in which a L(ow) tone triggers raising of a previous H(igh) to T(op), across a morpheme boundary in certain syntactic environments. No other combination of tones is seen to produce such a tonal change in this language. Crucially, leftward spreading of a L tone to H can in no obvious way account for an increase in the tonal level of a H tone since L has no apparent features whose spreading to H would increase its level. None of the currently available featural representations of tone (Wang 1967; Stevens and Halle 1971; Yip 1980; Pulleyblank 1986; Hyman 1979, 1993; Snider and van der Hulst 2012; Mamadou 2023, among many others) can straightforwardly capture this H-raising process, simply because Baatonum has four lexical level tones, which limits the representation a given tone may be assigned, within the available tonal feature models.

Nonetheless, we adopt the featural representation of tones proposed in Mamadou (2023) in

which the primitive features are -1 and +1, and where a given tone either has a simplex geometry with the potential for accommodating a single feature, or a complex geometry with the potential to maximally accommodate two features. The level of each tone is evaluated by adding up its features. Such a configuration yields the following five level tones: {+1,+1}, +1, {+1,-1}, -1, {-1,-1}, respectively evaluating to 2, 1, 0, -1, and -2 at the tonal level, assuming featural additivity. All tonal processes are reduced to the spreading operation under this model. However, this paper departs from Mamadou (2023) in the sense that discrete primitive features are not expressly provided. Instead, these discrete features emerge from raw inputs of continuous values. Furthermore, we assume that the four lexical tones of Baatonum evaluate to 2, 1, 0, and -1, respectively from highest to lowest, omitting -2, which does not occur lexically in Baatonum.¹

We shall show that the tonal interactions in Baatonum can be captured at the interpretable sub-symbolic neural level in the framework of Gradient Symbolic Computation (Smolensky and Goldrick 2016b, henceforth S&G or GSC), where discrete symbol structures – in this case, tonal levels – have a directly corresponding neural representation. Our account also adopts the concept of Tonal Complexes (Akinlabi and Liberman, 2001, 2007) in which tones are combined in a structured manner, such that constraints referring to such structured combinations can motivate a variety of tonal processes.

Of particular interest in this paper is a process of H tone raising (henceforth H-raising) to T when followed by a L tone across a morpheme boundary (H+L) in Baatonum. The crux of our analysis is that the H+L combination creates a tonal complex (henceforth ‘TC’), where, at the neural level, con-

¹In Mamadou (2023), the -2 was used to represent the intonational boundary tone, which is superlow.

nections between neurons representing tonal levels, the tonal complex, and input tonal features have the highest Harmony in the network when the T+L tonal level combination occurs. The difference in Harmony between the occurrence of a T+L combination and a H+L combination is enough to make the occurrence of a T before L more Harmonic than a H before L, even when inputs are such that a H would otherwise surface as the first tone. Because connection weights at the neural level can be positive or negative, the occurrence of a L-toned neuron represented as a -1 feature with a negative connection weight to a TC neuron will result a positive contribution to the overall Harmony. This is because in the GSC framework, the Harmony contribution of a connection between node n_1 with activation a_{n_1} and node n_2 with activation a_{n_2} and with connection weight w is the 3-way product of the two activations and the connection weight: $a_{n_1} * w * a_{n_2}$. It is the negative connection weight between a L-tone-level neuron and the TC that allows a L tone to cause a previous H tone to raise to Top. The following diagram shows the portion of the proposed network in which a L tonal level with activation -1 connects to a TC node, in a candidate where the TC node has a value of 1 as opposed to 0. Following S&G, we assume Quantization to have occurred for representations in the output, where a TC can only have a value of 0 or 1. The tl node below represents tonal level.

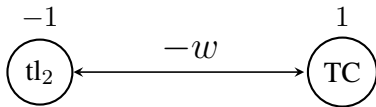


Figure 1: A network connection between two nodes

If a $-w$ connection weight represents a negative value and if tl_2 is quantized to -1 and TC to $+1$ then the Harmony of that portion of the network will be $+w$, a positive value.

The rest of the paper is organized as follows. In §2 we present the data used in our analysis. In §3 we introduce the GSC framework. In §4 we describe the quantization constraint of GSC that requires discrete, interpretable outputs. In §5 we discuss how weights and activations of the network were learned in order for the model to correctly predict all possible tonal patterns from learned input values. In §6 we show how H-raising is predicted by the network. We further illustrate this in §7 with example Harmonic tableaux that compare candi-

dates for some selected derivations. We conclude in §8 with a discussion of our findings.

2 Background and Data

The empirical data is from Baatonum, a language with four lexical level tones (T, H, M, and L, as illustrated in the quadruplet in (1)), and one intonational boundary tone. The process of interest was originally described by Welmers (1952) as one where a morpheme-final H tone raises to T when preceding a L tone across a morpheme boundary as shown in (2) below. However, Welmers' generalization left out the crucial fact that H-raising has additional morpho-syntactic conditioning. That is, the process does not occur, among other contexts, between the H tone of a direct object and the L tone of an indirect object (as in (3)), nor does it occur when the triggering L tone is part of a melody with more than one tone as shown in (4). In light of these new findings, we restate the Baatonum H-raising generalization as follows: a morpheme-final H tone raises to T when preceding a L tone across a morpheme boundary in certain syntactic constructions and when the conditioning L tone is the only tone in its melody.² For our purposes in this paper, however, the syntactic conditioning will be overlooked and we will only focus on the phonological conditioning of H-raising.

- (1) Baatonum tone quadruplet³
- a. nǎ̃ 'TOP'
 - b. nǎ̃ '1SG.NOM'
 - c. nǎ̃ 'come'
 - d. nǎ̃ '1SG.COND'

Another aspect of Baatonum tones relevant for our discussion in §8, is the restricted distribution of T, which never surfaces on verbs, and as a consequence is more restricted in sentence-final positions where verbs typically occur (due to the SOV word order). In fact, a test of frequency on a dataset with about 440 Baatonum sentences shows that the T tone has the lowest frequency of occurrence morpheme-initially, where it only represents 6% of all tones in this position.⁴

²The data used throughout this paper is from original fieldwork.

³The abbreviated glosses are as follows: TOP=Topic marker; NOM=Nominative; COND=Conditional; SG=Singular; PL=Plural; ACC=Accusative; IA=Inanimate. As far as tone notation goes, ǎ̃=Top; ǎ̃=High; ǎ̃=Mid; ǎ̃=Low.

⁴This is almost five times less than the frequency of M and L tones (at 26% each), and almost seven times less than that

(2) **H-raising (with /ná/ and /wá/)**

nǎ jè wǎ kò
1SG.NOM 3SG.Ina see already

"I've seen it."

(3) **No Raising: syntactic restriction**

ná núŋ jè wē
1SG.NOM 2SG.ACC 3SG.IA give

"I gave it to you."

(4) **No Raising: melodic restriction**

ú bòó mǒ
3SG.NOM pot have

"he has a pot."

ũ bòò mǒ
3SG.NOM wound have

"he has a wound."

3 The GSC framework

GSC is a research program in which symbol structures are derived by gradient representations and weighted constraints. Consisting of both a symbolic level of discrete structures and a sub-symbolic neural level, it bridges the gap between 'mind-driven' and 'brain-driven' approaches to studying human language. Recent work in this framework such as the following has spanned research in phonology, morphology, syntax and semantics: (Cho, 2019; Faust and Smolensky, 2017; Faust, 2017; Goldrick et al., 2016; Hsu, 2018; Müller, 2017; Rosen, 2016, 2018, 2019; Smolensky et al., 2014a; Smolensky and Goldrick, 2016a; van Hell et al., 2016; Zimmermann, 2017b,a, 2018). See also Hsu (2022) and references cited therein.

GSC has its origins in Legendre et al. (1990). Apart from this original proposal, Smolensky et al. (2014b), or other formative works such as Smolensky (1987), to our knowledge, none of the other above-mentioned research in the framework has explicitly made reference to its neural, sub-symbolic level. The fact that this study refers directly to the neural level is one of its main contributions, since it makes neural network implementations an interpretable way of addressing linguistic issues.

In our implementation of this framework, we abstract away from segmental structure such as vowels that bear tones and focus on tone melodies

of H (at 40%).

as autosegments (Leben, 1973; Goldsmith, 1976). A tone has a raw input that can have any activation value. As discussed in §1, at the featural level, a tone is given two separate discrete feature value nodes that are quantized to -1, 0 or +1. Tone levels are represented by a single node that is quantized to discrete values that can potentially be -1 (L), 0 (M), +1 (H) or +2 (T). Connections occur between the raw input nodes and the featural nodes, and between the featural nodes and the tone level nodes. There is also a negatively-weighted connection of each non-input node to itself that prevents its activation from blowing up to an uncontrolled value.⁵ These negative connection weights also act as Markedness constraints on the linguistic object represented by the node. For example, a negative self-connection on the node for tone level acts as a constraint against a tone level with a high absolute value. If the weight is $-w$, tone level 2 will incur a penalty of $2 * (-w) * 2 = -4w$ whereas tone level 1's penalty will only be $1 * (-w) * 1 = -w$. The greater Markedness of superhigh tones of level 2 is evident in their low frequency seen in elicitations as discussed above on page 2. A Mid tone with tone level 0 will incur no markedness penalty since $0 * (-w) * 0 = 0$. Thus, our model predicts that a M tone will act as a default tone.

It is important to note that unlike most networks used in Natural Language Processing, this is not a feed-forward network. Activation flows both ways in all node connections, even though we think of them in terms of input and output nodes.⁶ Instead, a learning process learns weights and activations that produce the highest Harmony for a given set of candidate outputs. The following diagram shows the proposed neural architecture for a single tone.

inp₁ and inp₂ are the two raw inputs with learned, continuous values, f₁ and f₂ the two feature values and tl the tone level. u₁ and u₂ are self-connections that produce 'unit-harmony'. Rather than fixing the raw input weights at discrete values, we gave the model the opportunity for these to be learned continuous values, since there are potential situations in the language where there may be lexical differences in whether words participate in tone raising that could be captured by differing raw input val-

⁵The self-connection is referred to as 'unit harmony' in the original GSC proposal.

⁶In the original GSC proposal, neural representations occur in a distributed fashion. Here, for ease of exposition, we use just a single node rather than a distributed set to represent a linguistic object.

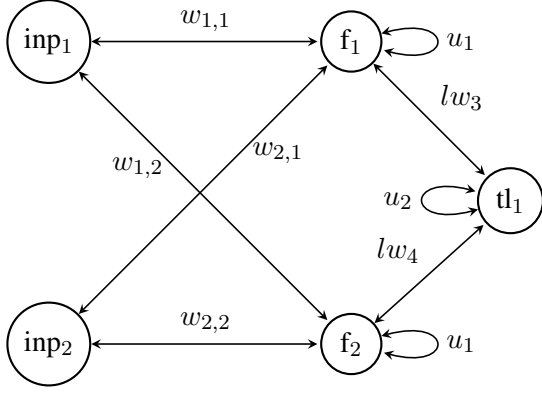


Figure 2: Neural architecture for a single tone

ues.

In this framework, the optimal output for a given input is the output candidate for which the Harmony of the whole network is the highest. Weighted constraints are expressed by connection weights between nodes.

4 Quantization

Quantization is an important part of the GSC framework. It involves a highly-weighted constraint that penalizes non-discrete outputs. We adapt from [Tupper et al. \(2018\)](#), the following two quantization constraints:

- **Feature level:** $H_{Q_1} = -[x^2(1-x^2)^2]$ is zero at values $+1$, 0 or -1 and negative otherwise.
- **Tone level:** $H_{Q_2} = -(x^2)(1-x^2)^2[1 - (\frac{x}{2})]^2$ is zero at values $+2$, $+1$, 0 , or -1 and negative otherwise.

The learning of weights and activations proceeds at an inverse rate to the enforcement of quantization constraints, where the effects of the former are gradually diminished and those of the latter gradually increased. (See [Smolensky et al. \(2014a\)](#) for details.)

5 Learning weights and activations

We ran a learning algorithm for weights and activations using Pytorch, Adam optimization ([Kingma and Ba, 2014](#)), a learning weight of 0.003, and a Cross-entropy loss function. All weights and activations were randomly initialized. For each of 8 possible tonal feature combinations that can be conceived to occur in the data – (1,1) T, (1,0) H, (1,-1) M, (0,1) H, (0,0) M, (0,-1) L, (-1,1) M, (-1,0) L – the model considers 72 possible candidates:

the power set of 9 tonal feature combinations (including $(-1,-1)$ which does not occur lexically), 4 tone levels: 2, 1, 0, -1 and two possible values for a tonal complex: 1 for present and 0 for absent. The model keeps the connection weights the same for each target output but learns different raw input activations for each target. Tables 1 and 2 below show, for a single tone, the activations and weights respectively that were learned by the model in one run of 145 iterations.

raw input activations		
Tonal features	inp1	inp2
(1,1)	1.310	0.298
(1,0)	2.066	-0.531
(1,-1)	1.423	-3.105
(0,1)	0.475	-2.501
(0,0)	0.0	0.0
(0,-1)	-0.47	-2.485
(-1,1)	-1.584	1.828
(-1,0)	-2.304	0.505

Table 1: Learned input values

Connection weights		
	$w_{,1}$	$w_{,2}$
Raw input 1 to features	2.626	-0.196
Raw input 2 to features	-1.013	1.802
Features to tone level	1.696	1.862
Feature self connection	-2.297	
Tone level self connection	-1.129	

Table 2: Learned connection weights

Figure 3 shows the above learned weights and activations for raw inputs, features and tone level, with learned input activations for feature set (1,0) and tone level 1 and a candidate with those quantized values.

Having one negatively activated input unit in the case of (1,0) features and both positive and negative connection weights makes it possible to derive a variety of different feature combinations and tone levels for different inputs.

Because weights and activations are initialized randomly for each run of the model, we will get somewhat different values each time. On this run, connections of each raw input node to its corresponding feature node are both positive whereas the “cross-connections” are negative. This means

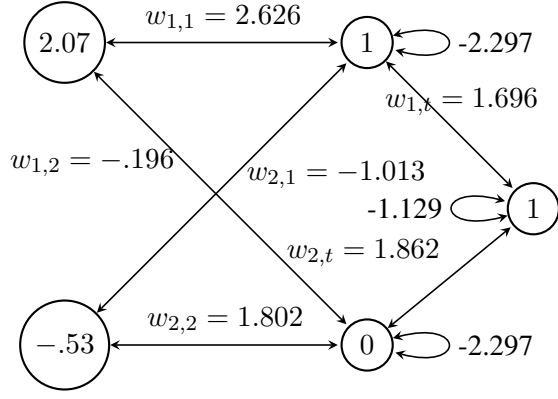


Figure 3: Learned weights and activations situated on the network

that very roughly speaking, the positive raw input in inp_1 will tend to result in a quantized 1 being optimal for feature +1. On the other hand, a negative raw input in inp_2 with a positive connection weight to feature 2 would result in negative Harmony for a candidate for a quantized +1 for the second tonal feature. And the negative cross-connection weights will also result in a positive Harmony contribution if +1 surfaces for feature 1 but negative Harmony if +1 surfaces in feature 2.

A (1,1) candidate fares worse, with an additional $(2.07 * (-.196)) + (-.53 * (-1.802)) = -1.36$ of additional negative Harmony. Added to this is the increased negative unit Harmony of self connections in the two feature nodes and the tone level node $(-2.297 * 2 - 1.129 * 2$ when the tonal level surfaces as 2).

6 Deriving H-raising

We look now at how the network derives raising of H to T when a L tone follows across a morpheme boundary.⁷ The following diagram shows two consecutive tones with a potential tonal complex node TC between them and connections from each tone level node to a tonal complex node. If the tonal complex node has a quantized value of 1, we consider there to be a tonal complex; otherwise it will be zero with no tonal complex.

We focus on the connection weights between tone levels tl_2 & tl_1 , and the tonal complex node TC, labeled respectively $-a$, b and c , where c is the self-connection of the TC node. Compare the following possible quantized output candidates:

⁷The morpheme boundary will not explicitly be represented in the network, but will be assumed to be present. We group it with the conditioning environments of H-raising (e.g. the syntactic condition) not taken into account in our analysis.

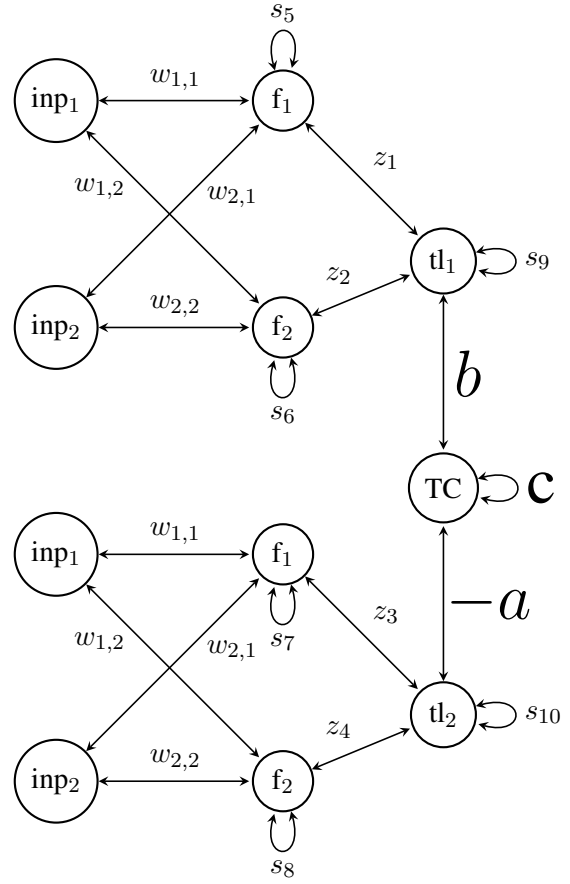


Figure 4: Network for two adjacent tones across a morpheme boundary

1. $\text{tl}_1 = 1$ (H), $\text{TC} = 1$, $\text{tl}_2 = -1$ (L) Harmony $= a + b - c$
2. $\text{tl}_1 = 1$ (H), $\text{TC} = 0$, $\text{tl}_2 = -1$ (L) Harmony $= 0$
3. $\text{tl}_1 = 2$ (T), $\text{TC} = 1$, $\text{tl}_2 = -1$ (L) Harmony $= 2b + a - c$
4. $\text{tl}_1 = 2$ (T), $\text{TC} = 0$, $\text{tl}_2 = -1$ (L) Harmony $= 0$

For the network to derive tone raising, it should be the case that the Harmony of this part of the network is such that it is greater for candidate 3 than candidate 4 (TC occurs for T+L), greater for candidate 2 than candidate 1 (no TC with H+L) and candidate 1 has greater Harmony than candidate 4 by a margin – call it m – greater than the margin by which a candidate with a H tone beats one with a T tone for an input that would derive a H tone. In other words, the network finds connection weights that make it more Harmonic for a T to occur than a H when a L tone follows it and a TC is formed.

The results of training for single tones were such that, for an input that derives a H tone, the Harmony for (1,0) (H) exceeds the Harmony for (1,1) (T) by 1.615 and the Harmony for (0,1) (H) exceeds the Harmony for (1,1) (T) by 1.704. So m must be greater than 1.704.

We can express the above conditions in terms of inequalities.

$$b + a < c$$

$$2b + a > c$$

$$2b - c + a > 1.704$$

We also want it to be the case that we don't get a TC if the second tone is not L. The best alternative to L that could try to derive a TC is a M tone whose $tl_2 = 0$, since a higher tone than that will produce negative Harmony with a negative connection weight on $-a$. If $tl_2 = 0$, then the harmony of this part of the network will be $2b - c$ if there is a T tone in first position. So we want $2b - c < 0$ or $2b < c$.

If we put all these conditions together and train this part of the network on them, making sure that the inequalities are satisfied by a further margin of at least 0.2, we get the following weights.

$$a = b = 2.11$$

$$c = 4.42$$

$$\text{So } a + b = 4.22 < c \text{ by } 0.2$$

$$2b + a = 6.33 > c$$

$$2b = 4.22 < c \text{ by } 0.2$$

$$2b - c + a = 1.91 \text{ which is greater than } 1.70 \text{ by } 0.21$$

A negative connection weight between the tone level node in tone 2 to a tonal complex node expresses a condition by the grammar to have the lowest possible tone as the rightmost element of a TC. Conversely, a positive connection weight from the first tone in the sequence to the TC node expresses a condition to have the highest possible tone level on the leftmost element. In other words, both conditions collectively maximize the distance or dissimilation between the two tonal members of the TC.

7 Example tableaux

There is not space to show a Harmonic tableau for every possible candidate for every input combination, but the following tableaux show derivations of a single H tone (Table 3), a single L tone (Table 4) and a H+L sequence (Table 5), where the input that would otherwise result in a H tone, surfaces as

T before a L tone and across a morpheme boundary. Because there are 9 sub-networks in the form of node-to-node connections for a single tone and 21 for a tone-boundary-tone sequence, we arrange the tableaux with constraints/sub-networks in rows and candidates in columns. Each candidate is represented by a bracketed pair of features followed by a tone level. For the H-tone tableau we show the top 10 candidates and for the L-tone tableau the top 7 plus a (0, 0) 0 candidate, with both tableaux having two sections and the winning candidate in the top section.

learned input: (2.06, -0.53) for a single H tone e.g., <i>nắ</i> '1SG.NOM' in (1-b)					
	(1,1) 2	(1,1) 1	(1,0) 2	(1,0) 1	(1,0) 0
$i_1 \Leftrightarrow f_1$	5.424	5.424	5.424	5.424	5.424
$i_1 \Leftrightarrow f_2$	-0.405	-0.405	-0.0	-0.0	-0.0
$i_2 \Leftrightarrow f_1$	0.538	0.538	0.538	0.538	0.538
$i_2 \Leftrightarrow f_2$	-0.957	-0.957	-0.0	-0.0	-0.0
$f_1 \Leftrightarrow f_1$	-2.297	-2.297	-2.297	-2.297	-2.297
$f_2 \Leftrightarrow f_2$	-2.297	-2.297	-0.0	-0.0	-0.0
$tl \Leftrightarrow tl$	-4.517	-1.129	-4.517	-1.129	-0.0
$f_1 \Leftrightarrow tl$	3.392	1.696	3.392	1.696	0.0
$f_2 \Leftrightarrow tl$	3.724	1.862	0.0	0.0	0.0
tot _H	2.604	2.434	2.539	4.231	3.664
	(0,0) 0	(1,-1) -1	(1,-1) 0	(1,-1) 1	(1,0) -1
$i_1 \Leftrightarrow f_1$	0.0	5.424	5.424	5.424	5.424
$i_1 \Leftrightarrow f_2$	0.0	0.405	0.405	0.405	0.0
$i_2 \Leftrightarrow f_1$	0.0	0.538	0.538	0.538	0.538
$i_2 \Leftrightarrow f_2$	-0.0	0.957	0.957	0.957	-0.0
$f_1 \Leftrightarrow f_1$	-0.0	-2.297	-2.297	-2.297	-2.297
$f_2 \Leftrightarrow f_2$	-0.0	-2.297	-2.297	-2.297	-0.0
$tl \Leftrightarrow tl$	-0.0	-1.129	-0.0	-1.129	-1.129
$f_1 \Leftrightarrow tl$	0.0	-1.696	0.0	1.696	-1.696
$f_2 \Leftrightarrow tl$	0.0	1.862	-0.0	-1.862	-0.0
tot _H	0.0	1.765	2.729	1.433	0.839

Table 3: Tableau for H tone derivation

In the first tableau, a single H tone with tone level 1 (second-last column) has the highest total Harmony. It avoids negative Harmony for input 2 to feature 2 (4th row) and has less tone level unit Harmony penalty (row $tl \Leftrightarrow tl$) than candidates 1 and 3.

The next tableau shows harmonies for top-scoring candidates when the raw input has the learned values for tone (-1, 0) L. We see that the optimal candidate does have those tone features and in addition a tone level of -1.

In the last tableau we look at the case of an underlying H followed by and underlying L across

learned input: (-2.304, 0.505) for L tone e.g., nà '1SG.COND' in (1-d)				
	(-1,0) 0	(-1,0) -1	(-1,1) 0	(0,0) 0
$i_1 \Leftrightarrow f_1$	6.05	6.05	6.05	-0.0
$i_1 \Leftrightarrow f_2$	0.0	0.0	0.452	0.0
$i_2 \Leftrightarrow f_1$	0.511	0.511	0.511	-0.0
$i_2 \Leftrightarrow f_2$	0.0	0.0	0.909	0.0
$f_1 \Leftrightarrow f_1$	-2.297	-2.297	-2.297	-0.0
$f_2 \Leftrightarrow f_2$	-0.0	-0.0	-2.297	-0.0
$tl \Leftrightarrow tl$	-0.0	-1.129	-0.0	-0.0
$f_1 \Leftrightarrow tl$	-0.0	1.696	-0.0	0.0
$f_2 \Leftrightarrow tl$	0.0	-0.0	0.0	0.0
tot _H	4.264	4.831	3.328	0.0
	(-1,-1) -1	(-1,0) 1	(-1,1) -1	(-1,1) 1
$i_1 \Leftrightarrow f_1$	6.05	6.05	6.05	6.05
$i_1 \Leftrightarrow f_2$	-0.452	0.0	0.452	0.452
$i_2 \Leftrightarrow f_1$	0.512	0.512	0.512	0.512
$i_2 \Leftrightarrow f_2$	-0.91	0.0	0.91	0.91
$f_1 \Leftrightarrow f_1$	-2.297	-2.297	-2.297	-2.297
$f_2 \Leftrightarrow f_2$	-2.297	-0.0	-2.297	-2.297
$tl \Leftrightarrow tl$	-1.129	-1.129	-1.129	-1.129
$f_1 \Leftrightarrow tl$	1.696	-1.696	1.696	-1.696
$f_2 \Leftrightarrow tl$	1.862	0.0	-1.862	1.862
tot _H	3.033	1.439	2.034	2.366

Table 4: Tableau for derivation of L tone

a morpheme boundary in a syntactic environment that should produce tone raising of the H to T. Here, we focus on the three connections involved with the TC node that connects the networks of the two consecutive tones. The TC node is quantized to allow only values of 0 (no TC) or 1 (TC present). Our candidates are the power set of the candidates in each of the two previous tableaux, multiplied further by 2, for each of the possible values of the TC node. This gives us $4 \times 4 \times 2 = 32$ candidates and 3 subnetworks to consider – one for each of the connections involving the TC node: between tl_1 and TC, between tl_2 and TC and TC's self-connection. As was discussed above, the learned weights for these were 2.11, -2.11 and -4.42. For candidates, we look at all possible combinations of the 4 quantized values for the two tone levels and the 2 quantized values for the TC node.

Because we have so many candidates, in this tableau we revert to the conventional tableau format, with candidates in rows and constraints – in this case subnetworks – in columns. The candidates in the left column of each row are in the format tl_1 , tl_2 and tc . The next 3 columns give the Harmony of each of the 3 subnetwork connections and each of the last two each give the highest Harmony among all possible featural candidates for the correspond-

ing tone level in the leftmost column. For example, since we are looking at inputs, which for single tones would derive a H and a L respectively, in the first row, t_1 is the highest Harmony of any possible candidate for the raw inputs values that were learned for a H tone that result in a tone level of +1. And in the same row, t_2 is the highest Harmony of any possible candidate for the raw inputs values that were learned for a L tone that result in a tone level of -1.

We can see that the candidate '2 -1 1' which indicates a tone level 2 (T) on the first tone and -1 (L) on the second is most optimal, even though the input for the first tone is such that a H would surface there in other contexts.

learned input: (2.06, -0.53) for a H tone followed by (-2.30, 0.51) for L tone across morpheme boundary e.g., wǎ kò 'see already' in (2)						
Cand	$tl_1 \Leftrightarrow tc$	$tl_2 \Leftrightarrow tc$	$tc \Leftrightarrow tc$	t_1	t_2	tot _H
-1 -1 0	0.0	-0.0	-0.0	1.76	4.83	6.6
-1 -1 1	-2.11	2.11	-4.42	1.76	4.83	2.18
-1 0 0	0.0	-0.0	-0.0	1.76	4.26	6.03
-1 0 1	-2.11	-0.0	-4.42	1.76	4.26	-0.5
-1 1 0	0.0	-0.0	-0.0	1.76	2.37	4.13
-1 1 1	-2.11	-2.11	-4.42	1.76	2.37	-4.51
-1 2 0	0.0	-0.0	-0.0	1.76	-0.86	0.91
-1 2 1	-2.11	-4.22	-4.42	1.76	-0.86	-9.84
0 -1 0	0.0	-0.0	-0.0	3.66	4.83	8.5
0 -1 1	0.0	2.11	-4.42	3.66	4.83	6.19
0 0 0	0.0	-0.0	-0.0	3.66	4.26	7.93
0 0 1	0.0	-0.0	-4.42	3.66	4.26	3.51
0 1 0	0.0	-0.0	-0.0	3.66	2.37	6.03
0 1 1	0.0	-2.11	-4.42	3.66	2.37	-0.5
0 2 0	0.0	-0.0	-0.0	3.66	-0.86	2.81
0 2 1	0.0	-4.22	-4.42	3.66	-0.86	-5.83
1 -1 0	0.0	-0.0	-0.0	4.23	4.83	9.06
1 -1 1	2.11	2.11	-4.42	4.23	4.83	8.86
1 0 0	0.0	-0.0	-0.0	4.23	4.26	8.5
1 0 1	2.11	-0.0	-4.42	4.23	4.26	6.19
1 1 0	0.0	-0.0	-0.0	4.23	2.37	6.6
1 1 1	2.11	-2.11	-4.42	4.23	2.37	2.18
1 2 0	0.0	-0.0	-0.0	4.23	-0.86	3.38
1 2 1	2.11	-4.22	-4.42	4.23	-0.86	-3.15
2 -1 0	0.0	-0.0	-0.0	2.6	4.83	7.44
2 -1 1	4.22	2.11	-4.42	2.6	4.83	9.35
2 0 0	0.0	-0.0	-0.0	2.6	4.26	6.87
2 0 1	4.22	-0.0	-4.42	2.6	4.26	6.67
2 1 0	0.0	-0.0	-0.0	2.6	2.37	4.97
2 1 1	4.22	-2.11	-4.42	2.6	2.37	2.66
2 2 0	0.0	-0.0	-0.0	2.6	-0.86	1.75
2 2 1	4.22	-4.22	-4.42	2.6	-0.86	-2.67

Table 5: Tableau for H raising

8 Discussion and Conclusion

The results of this study lend support to empirical evidence in favor of a featural representation of tone (McPherson 2017; Meyase 2021, but see Hyman 2010; Clements et al. 2010 for an opposite view on this), from a representational standpoint. More specifically, the finding in this study that feature-based learning of phonological processes is possible, lends additional support for this type of representation, highlighting the versatility of the type of numerical tone features introduced in Mamadou (2023). At a computational level, we show that Gradient Symbolic Computation offers directly interpretable neural level connections, where markedness constraints that penalize illicit surface forms can be straightforwardly read off of the connection weights. This direct interpretability makes GSC appropriate for both discrete and gradient linguistic modeling.

The gradient modeling capability of GSC can be seen in its ability to model what we might call *gradient dissimilation*. That is, H-tone raising seeks to maximize the difference in tone levels across a morpheme boundary when it raises a H tone to a T tone, and there is a threshold at which the tone level difference must occur – in this case, a level difference of $2 - (-1) = 3$, which is the maximum distance between the lowest and the highest lexical tones in Baatonum. This threshold is determined by the connection weights of the network, and if these weights were different, it would be possible, for example in another tone language, to have this kind of dissimilation occur with different tone levels involved. In other words, the typology of tonal processes may emerge from slight changes in the gradient connection weights of the network. We plan in further research to explore these kinds of possibilities in other tone languages.

Also noteworthy is how the network captured distributional restrictions of some tones. As shown on page (1), the Top tone is generally marked in Baatonum as evidenced in its restricted distribution frequency of only 6% of all morpheme initial tones; though this frequency rises to 11% in morpheme-final positions, due in part to the H-raising in that position. The way the network captures this is by assigning a higher markedness penalty to the Top tone for having the highest tone level of 2, which quadruples the effect of the negative self-connection. (See calculations of these values in paragraph 3 of §3.) This suggests that in languages

with a crowded tonal space like Baatonum, the tones at the extremas will be most marked (recall the markedness penalty of $2 * (-w) * 2 = -4w$ for tone level 2, whereas tone level 1's penalty will only be $1 * (-w) * 1 = -w$, on page 3). This same constraint is what makes tone level 0 (M) the least marked tone in the language.

We showed this in an experiment where we ran 360,000 iterations in which raw input values ranged at 0.1 increments from -3 to 3 and weights were given random values on a uniform distribution from -3 to -0.5 for self-connections, 0 to 3 for connections from features to tone level and -3 to 3 for other connections. Separate runs yielded almost identical results for the frequency at which a tone level occurred. Mid tones were the most frequent at 48.2% and T tones the least frequent at 7.9%. These results are consistent with our model's predictions that M tone should be considered a default tone and T as the most marked tone.

A final point about our results here is the high total harmony (tot_H) of some candidates in the H-raising tableau in Table 5. Consider the following three candidates with the highest harmony: the winning candidate with a Top tone followed by a Low tone (across morpheme boundary) within a tonal complex (i.e. cand (2,-1,1)), the losing candidate with the next highest harmony (1,-1,0), and the one with the third highest harmony, (1,-1,1). The first thing to note is that the candidate with T+L within a TC will, unsurprisingly, always be the most harmonic candidate. However, note that it is more costly for a H+L to form a TC than it is for a H to simply precede a L across a morpheme boundary without any TC being formed around them. And the fact that a candidate like (1,-1,0) has the second highest harmony captures well the Baatonum facts about how a H can precede L within the same morpheme, and even across morpheme boundary in some syntactic environments. That is, the H+L sequence is licit enough to simply occur, but not enough to occur within a TC.

From the results and their discussion in this section, it is clear that GSC is not only an adequate, but also an ideal framework for linguistic modeling. Future work within this framework will explore typological implications of the connection weights in the learning of phonological processes in general, and tonal processes in particular.

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